

Logic Gates

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George Boole has written a book "An investigation of the laws of thought" this book is based on logic and Probabilities.

Later on Claude E. Shannon at Massachusetts Institute of Technology wrote a thesis. Today Boolean Algebra is the backbone of Computer Circuit Analysis.

1 and 0
High and low
True and False
On and Off

Fundamental concept of boolean algebra

Only two mathematical operations

addition

OR

Conjunction (Logical Addition)

The '+' symbol is read as X or Y

$$\begin{array}{lcl} X + Y & & \\ 0 + 0 & = & 0 \\ 0 + 1 & = & 1 \\ 1 + 0 & = & 1 \\ 1 + 1 & = & 1 \end{array}$$

Multiplication (Repeated addition)

AND

Disjunction (Logical multiplication)

$$\begin{array}{lcl} X \cdot Y & & \\ 0 \cdot 0 & = & 0 \\ 1 \cdot 0 & = & 0 \\ 0 \cdot 1 & = & 0 \\ 1 \cdot 1 & = & 1 \end{array}$$

$a > b$ & $a > c$ Then a is greatest

Three basic laws of Boolean Algebra are same as normal algebra:

1. Commutative law $\rightarrow A+B = B+A, A \cdot B = B \cdot A$
2. Associative law $\rightarrow (A+B)+C = A+(B+C), A(B \cdot C) = (A \cdot B) \cdot C$
3. Distributive law $\rightarrow A(B+C) = AB+AC$
 $A+BC = (A+B)(A+C)$

Logic Gates: A logic gate is defined as an electronic circuit with two or more input signals and it depicts the logic clearly.

Three types of Logic Gates:

1. Basic Gates

→ OR gate
→ AND gate
→ NOT gate

2. Universal Gates

→ NOR gate
→ NAND gate

3. Exclusive Gates

→ XOR gate
→ XNOR gate

→ AND gate
→ NOT gate

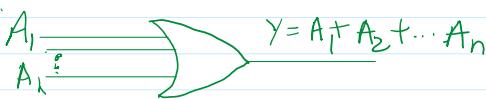
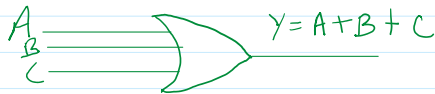
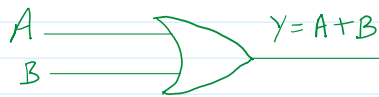
→ NAND gate

→ XNOR gate.

BASIC GATES:

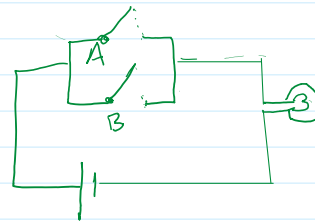
(i) OR Gate :- An OR gate is a logic ckt. with 2 or more input signals and '1' output signal. The output signal will be high or '1', if any of the input signal is high or '1', or true or on (logic-1).

OR gate performs logical Addition. It is represented by '+' plus symbol.



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

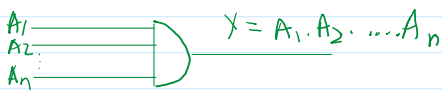
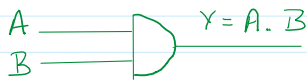
Truth table of OR gate.



Electrical representation of OR gate.

(ii) AND Gate :- An AND gate is a logic ckt. with 2 or more input signals and '1' output signal. The output signal will be high or '1', if all of the input signal is high or '1', or true or on (logic-1).

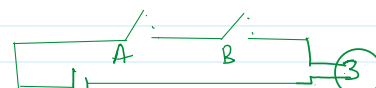
AND gate performs logical Multiplication. It is represented by '.' dot symbol.



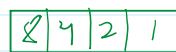
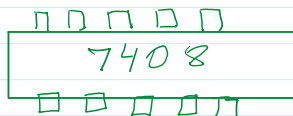
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Truth table of AND gate.

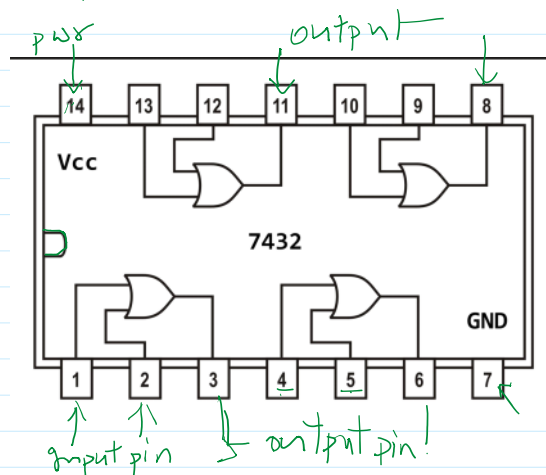
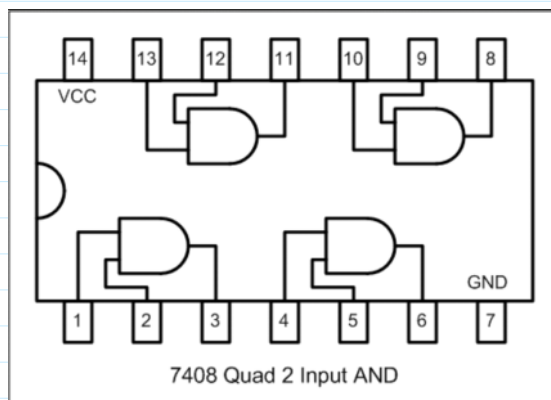
Integrated circuit (IC)



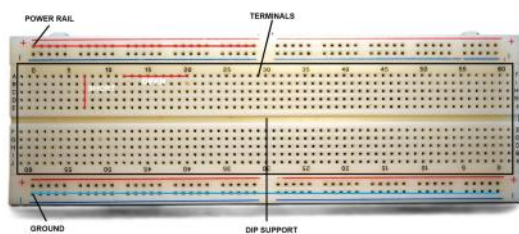
Integrated circuit (IC)



quad input → Registers
→ capacitors
→ diodes.
← 16 different combination



Bread board



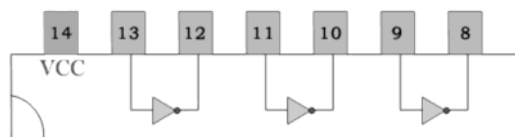
NOT Gate



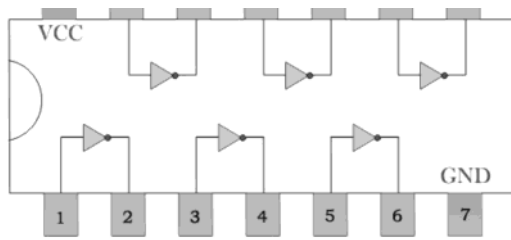
An inverter or NOT gate is the ckt. that takes one input signal and output signal is always opposite or complement of the input signal.

A	$\bar{A}$
0	1
1	0

Truth table for NOT gate.



Inverter ckt.



7404 Six Inverter

NOT gate IC.

Inverter ckt.
or
short ckt.

BASIC DUALITY IN BOOLEAN ALGEBRA

We state the duality theorem in the following

1. changing each OR (+) sign to an AND (·) sign.
2. changing each AND (·) sign to an OR (+) sign.
3. Complement each '0' to '1' or '1' to '0'

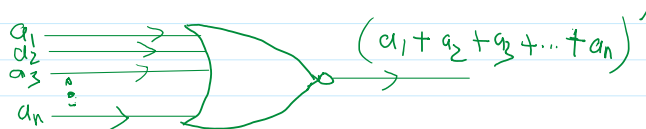
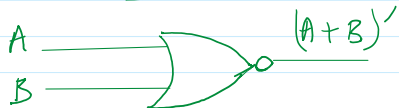
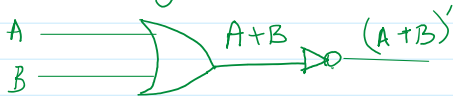
$$A + 0 = A \cdot 1 = A$$

dual relationship $\left\{ \begin{array}{l} A \cdot (B + C) = AB + AC \\ A + B \cdot C = (A + B)(A + C) \end{array} \right\}$ distributive law

$$\begin{aligned} \text{dual of } (1 + A) &= 0 \cdot A = 0 \\ (1 + \bar{A}) &= 0 \cdot \bar{A} = 0 \end{aligned}$$

Universal Gates (De-Morgan's law)

1. NOR: This gate takes two or more input and output is inverter of OR gate.

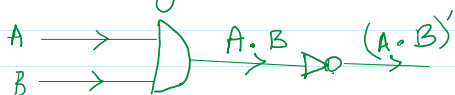


A	B	(A+B)'
0	0	1
0	1	0
1	0	0
1	1	0

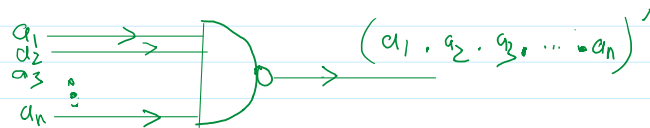
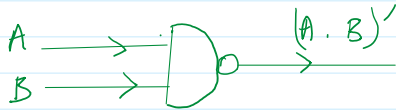
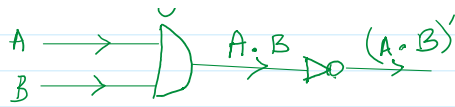
Truth table.

$$(A+B)' = \bar{A} \cdot \bar{B}$$

2. NAND: This gate takes two or more input and output is inverter of AND gate.



A	B	(A.B)'
0	0	1



A	B	(A.B)'
0	0	1
0	1	1
1	0	1
1	1	0

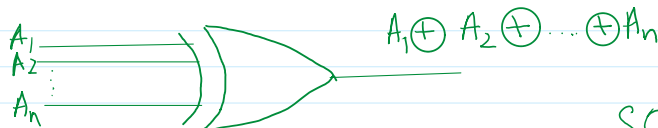
Truth table.

$$(A.B)' = \bar{A} + \bar{B}$$

EXCLUSIVE GATES

1. XOR Gate : It takes 2 or more input signals and output one signal. The output signal is high or 1 or true, iff we had odd number of high or 1 or true input signals, otherwise output is low or 0.

It is represented by '+' plus in a circle symbol.



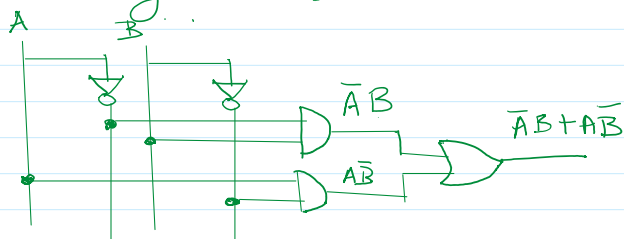
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	0

$A+B \leftarrow$
 $\checkmark \bar{A}B$
 $\checkmark A\bar{B}$
 $A+B \leftarrow$

product

SOP (sum of product) $\rightarrow A+B = \bar{A}B + A\bar{B}$

Drawing $\bar{A}B + A\bar{B}$ which is XOR gate using basic gates.



XOR Gate

$$A+B = (A+B) \cdot (\bar{A} + \bar{B})$$

POS (product of sum)

	A	B	C	A+B+C
0	0	0	0	0 ✓
1	0	0	1	1 ✓
2	0	1	0	1 ✓
3	0	1	1	0 ✓
4	1	0	0	1 ✓
5	1	0	1	0 ✓
6	1	1	0	0
7	1	1	1	1 ✓

SOP

$$A+B+C = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

POS $\rightarrow 1 \rightarrow$ is barred.

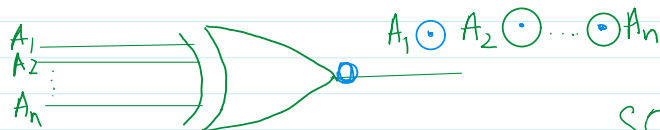
$$A+B+C = (A+B+C) \cdot (A+\bar{B}+\bar{C}) \cdot (\bar{A}+B+\bar{C}) \cdot (\bar{A}+\bar{B}+C)$$

Home work, draw the logic ckt. by using basic gates.

XNOR:-

1. **XNOR Gate** : It takes 2 or more input signals and output one signal. The output signal is high or 1 or true, iff we had even number of high or 1 or true input signals, otherwise output is low or 0.

It is represented by ' $\odot$ ' plus in a circle symbol.

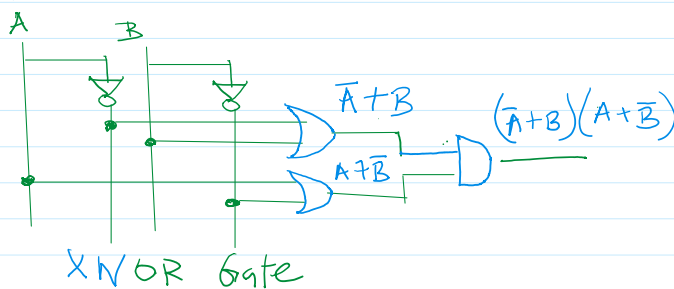


A	B	A ⊙ B
0	0	1
0	1	0 ✓
1	0	0 ✓
1	1	1

$\bar{A}\bar{B}$
 $(A+\bar{B})$
 $(\bar{A}+B)$ } POS
 AB

SOP (sum of product) $\rightarrow A \odot B = \bar{A}\bar{B} + A.B$

POS $A \odot B = (A+\bar{B})(\bar{A}+B)$



H.W Draw the ckt using basic gates.

	A	B	C	A ⊙ B ⊙ C
0	0	0	0	1
1	0	0	1	0 ✓
2	0	1	0	0 ✓
3	0	1	1	1
4	1	0	0	0 ✓
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0 ✓

$$G = A \odot B \odot C$$

SOP

$$G = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + ABC$$

POS

$$G = (A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)(\bar{A}+\bar{B}+\bar{C})$$

H.W Draw the logic ckt using basic gates.

Next class:- Application